

St. Xavier's Institution (Panihati)

SHORT TEST 2025-26

CLASS XI - Artificial Intelligence

Maximum Marks: 35

Time Allowed: $1\frac{1}{2}$ Hour

Instructions:

- Attempt all questions.
 - Marks are indicated in brackets next to each question.
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Section A

Question 1: Write the correct answer only.

[1 × 5 = 5 marks]

- a) Which of the following is an example of probabilistic computing?
(a) Time tables (b) Maps (c) Report cards (d) Weather forecast
- b) What is 'Broad AI' ?
(a) Strong AI (b) Weak AI (c) General AI (d) Both (a) and (c)
- c) The expression `a//b` in Python means:
(a) Floor division (b) Division (c) Power (d) Modulus
- d) Who is known as the Father of Artificial Intelligence?
(a) Alan Turing
(b) John McCarthy
(c) Josef Capek
(d) Joseph F. Engelberger
- e) Which Python structure allows data access but not modification (immutable)?
(a) List (b) Dictionary (c) Set (d) Tuple

Section B

Question 2: Answer the following in brief.

[1 × 10 = 10 marks]

- a) Differentiate between a square matrix and a rectangular matrix with examples.
- b) What are key roles of cybersecurity in AI?
- c) Determine the type of the following matrices: (write the answers only)

(I) $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(II) $B = [5]$ (bonus question)

(III) $C = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

(IV) $D = [1 \ 2 \ 3]$

- d) Write the output of the following code with explanation.

```
text = "Python World!"  
print(text[7:13])
```

- e) Write a Python program to remove duplicate elements from a list.
- f) How do you identify if a dataset is relevant and complete for your AI project?
- g) What is DATA? Write the name of different types of data?
- h) Write TWO common types of errors in Python.
- i) Write a two-by-two matrix with matrix elements $a_{ij} = i - j$.
- j) Write the truth table of AND gate.

Section C

(Long answer type question)

[4 × 5 = 20 marks]

Question 3

Write a program that takes a string as input and returns the count of vowels (a, e, i, o, u) in the string. [4]

Question 4

Explain the five different stages of AI project cycle. [4]

Question 5

- a) Compare Human Intelligence and Artificial Intelligence in at least FOUR key areas. [2]
- b) Write FOUR key milestone of AI. [2]

Question 6

- a) Explain NAND and NOR gates with truth tables and expressions. Why are these gates called universal gates? [4]

Question 7

- (a) Show that matrix multiplication is not commutative using:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Compare AB and BA . [2]

- (b) Let

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Find:

- i) $P \times P^T$
- ii) $P^T \times P$

Comment on the dimensions of the resulting matrices. [2]